(iii) Find the height of a person whose forehand length is 40 cm.

(iv) Find the length of forehand of a person if the height is 53.3 inches.

1.6 Representation of Functions

A function may be represented by

(a) a set of ordered pairs (b) a table form

(c) an arrow diagram (d) a graphical form

Let $f : A \rightarrow B$ be a function

(a) Set of ordered pairs

The set $f = \{(x, y) \mid y = f(x), x \in A\}$ of all ordered pairs represent a function.

(b) Table form

The values of x and the values of their respective images under f can be given in the form of a table.

(c) Arrow diagram

An arrow diagram indicates the elements of the domain of f and their respective images by means of arrows.

(d) Graph

The ordered pairs in the collection $f = \{(x, y) \mid y = f(x), x \in A\}$ are plotted as points in the XY- plane. The graph of f is the totality of all such points.

Every function can be represented by a curve in a graph. But not every curve drawn in a graph will represent a function.

The following test will help us in determining whether a given curve is a function or not.

1.6.1 Vertical line test

A curve drawn in a graph represents a function, if every vertical line intersects the curve in at most one point.

Example 1.10 Using vertical line test, determine which of the following curves (Fig.1.18(a), 1.18(b), 1.18(c), 1.18(d)) represent a function?

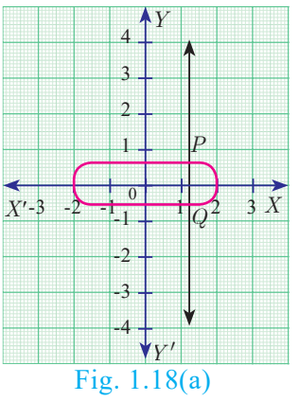


Fig. 1.18(a)

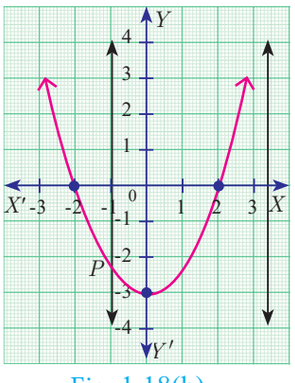


Fig. 1.18(b)

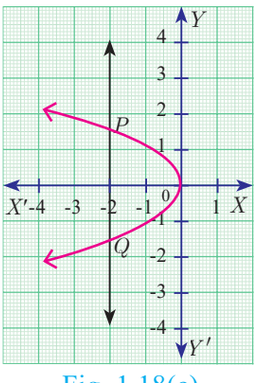


Fig. 1.18(c)

1 Relation and Functions.indd 15 15-12-2020 02:50:14 PM